PriorRG: Prior-Guided Contrastive Pre-training and Coarse-to-Fine Decoding for Chest X-ray Report Generation

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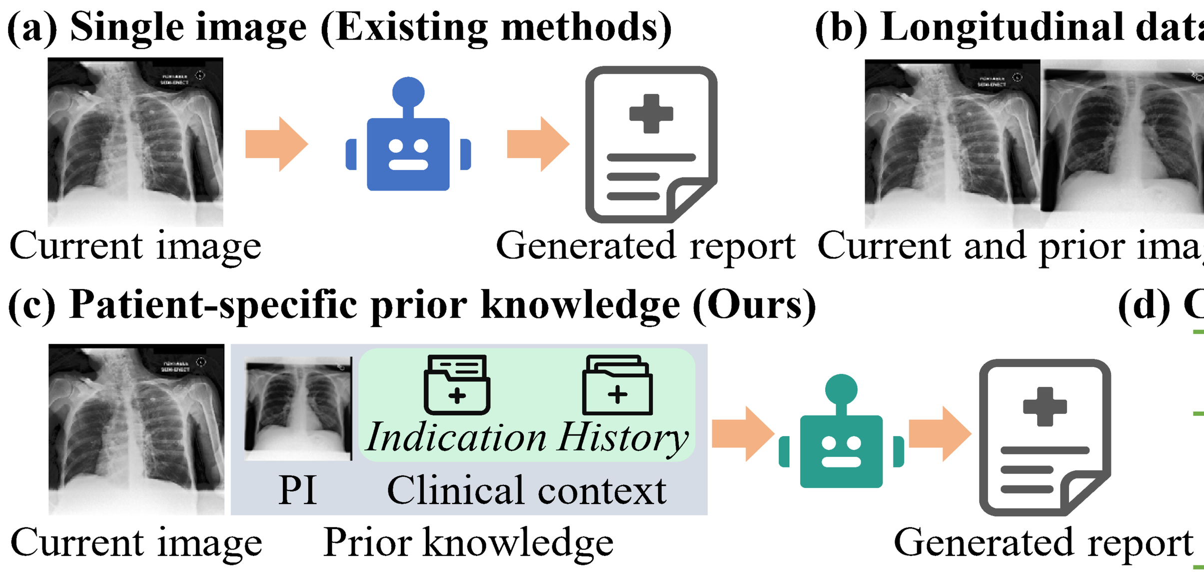
JANUARY 20–27, 2026 | SINGAPORE

Introduction

- **Background:** Chest X-ray report generation aims to reduce radiologists' workload by automatically producing high-quality preliminary reports.
- **Contribution:** ① We propose **PriorRG**, which integrates **patient-specific prior knowledge** to generate **context-aware and disease progression-oriented reports**. ② We introduce **a prior-guided contrastive pre-training scheme** that mirrors diagnostic reasoning by using clinical context to guide spatiotemporal features extraction, thereby improving alignment with report semantics and boosting medical image-text retrieval performance. ③ We present **a prior-aware coarse-to-fine decoding** that incrementally integrates clinical context, disease progression patterns, and hierarchical visual cues, enhancing the clinical accuracy and fluency of generated reports.
- **Results:** Extensive experiments on MIMIC-CXR and MIMIC-ABN datasets demonstrate that PriorRG outperforms state-of-the-art methods, achieving a **3.6% BLEU-4** and **3.8% F1 score** improvement on MIMIC-CXR**,** and a **5.9% BLEU-1** gain on MIMIC-ABN**.** Furthermore, PriorRG is computationally efficient, with an average inference time of **0.1245 seconds per sample** and a GPU memory footprint of **7.68 GB**.

Motivation

PriorRG jointly models disease progression (DP) and clinical context (CC) to enhance cross-modal alignment and improve the quality of generated reports.



(d) Comparison of characteristics

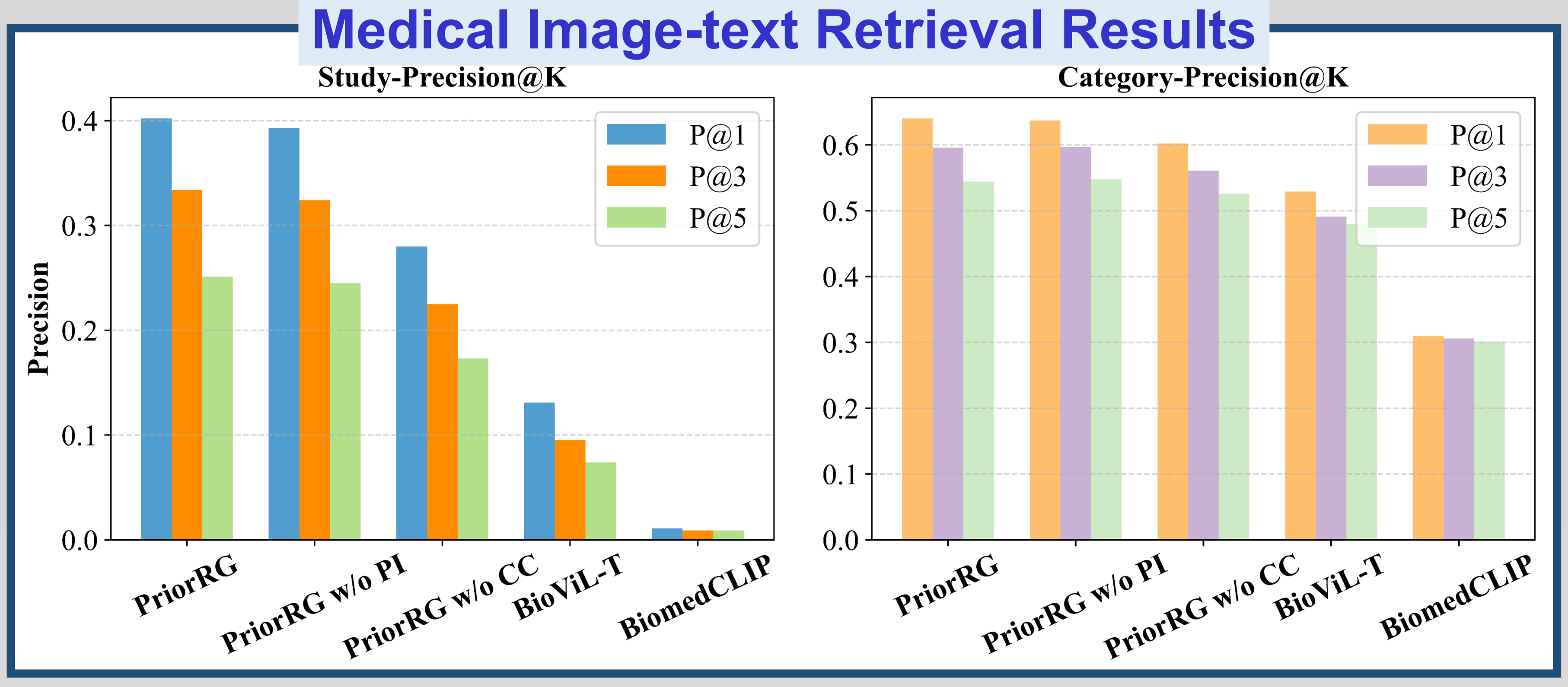
Trait	(a)	(b)	Ours
CC	×	×	✓
DP	×	✓	✓

Report Generation Performance

Dataset	Method	Venue	NLG Metrics ↑ B-1	B-2	B-3	B-4	MTR	R-L	CE Metrics ↑ P	R	F1
M-CXR	KiUT	CVPR'23	0.393	0.243	0.159	0.113	0.160	0.285	0.371	0.318	0.321
	METransformer	CVPR'23	0.386	0.250	0.169	0.124	0.152	0.291	0.364	0.309	0.311
	CoFE	ECCV'24	-	-	-	0.125	<u>0.176</u>	0.304	0.489	0.370	0.405
	DCG	MM'24	0.397	0.258	0.166	0.126	0.162	0.295	0.441	0.414	0.404
	MAN	AAAI'24	0.396	0.244	0.162	0.115	0.151	0.274	0.411	0.398	0.389
	R2GenGPT	Meta-Radio'23	0.411	0.267	0.186	0.134	0.160	0.297	0.392	0.387	0.389
	Med-LLM	MM'24	-	-	-	0.128	0.161	0.289	0.412	0.373	0.395
	R2-LLM	AAAI'24	0.402	0.262	0.180	0.128	0.175	0.291	0.465	<u>0.482</u>	<u>0.473</u>
	SEI	MICCAI'24	0.382	0.247	0.177	0.135	0.158	0.299	<u>0.523</u>	0.410	0.460
	HERGen	ECCV'24	0.395	0.248	0.169	0.122	0.156	0.285	-	-	-
	MPO	AAAI'25	**0.416**	<u>0.269</u>	<u>0.191</u>	<u>0.139</u>	0.162	<u>0.309</u>	0.436	0.376	0.353
	PriorRG (Ours)	-	<u>0.412</u>	**0.290**	**0.220**	**0.175**	**0.189**	**0.324**	**0.541**	**0.485**	**0.511**
	△(%) ↑	-	-0.4	+2.1	+2.9	+3.6	+1.3	+1.5	+1.8	+0.3	+3.8
M-ABN	CMN◊	ACL'21	0.256	0.147	0.095	0.066	0.110	0.230	<u>0.466</u>	<u>0.454</u>	<u>0.460</u>
	SEI◊	MICCAI'24	<u>0.267</u>	<u>0.157</u>	<u>0.104</u>	<u>0.073</u>	<u>0.114</u>	<u>0.231</u>	<u>0.466</u>	0.408	0.435
	PriorRG (Ours)	-	**0.326**	**0.201**	**0.139**	**0.102**	**0.140**	**0.242**	**0.467**	**0.476**	**0.471**
	△(%) ↑	-	+5.9	+4.4	+3.5	+2.9	+2.6	+1.1	+0.1	+2.2	+1.1

Table 1: Comparison of report generation performance with SOTA methods on MIMIC-CXR (M-CXR) and MIMIC-ABN (M-ABN) datasets. △ indicates the performance difference between PriorRG and the best baseline. ◊ denotes reproduced results; others are cited from the original papers. The **best** and <u>second-best</u> values are in **bold** and <u>underlined</u>, respectively.

Medical Image-text Retrieval Results

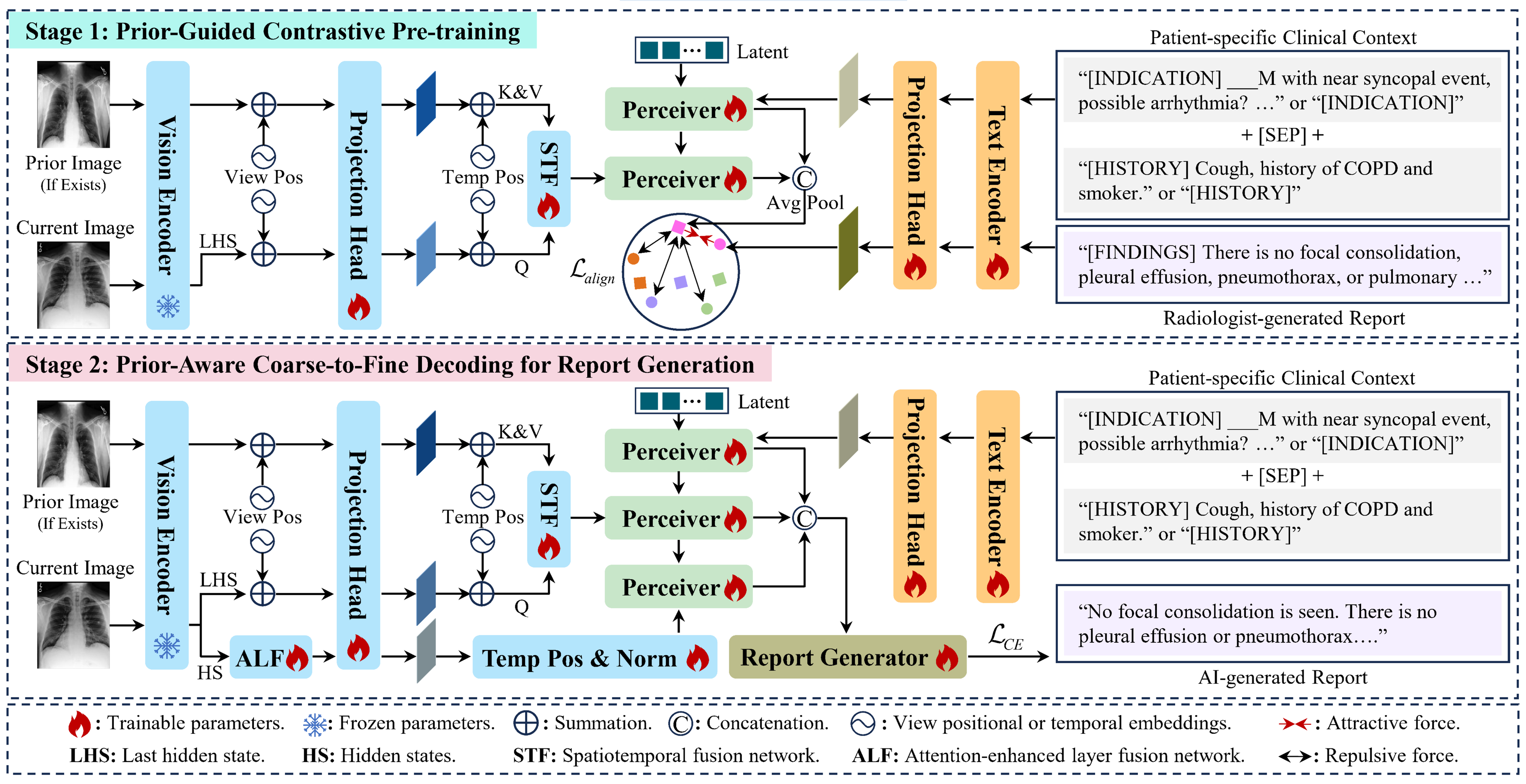


Ablation Study of Report Generation on MIMIC-CXR

Model	Stage 1	Stage 2 CC	PI	Hidden States	NLG Metrics ↑ B-1	B-2	B-3	B-4	MTR	R-L	CE Metrics ↑ P	R	F1
(a)	✓	✗	✗	✗	0.355	0.219	0.149	0.108	0.154	0.262	0.521	0.431	0.472
(b)	✓	✗	✗	✓	0.357	0.216	0.143	0.102	0.154	0.255	0.513	0.480	0.496
(c)	✓	✓	✗	✗	0.405	0.283	0.214	0.170	0.186	0.320	0.517	0.460	0.487
(d)	✓	✓	✗	✓	0.404	0.285	0.217	0.173	0.187	**0.328**	0.540	0.477	0.507
(e)	✓	✓	✓	✗	0.400	0.282	0.214	0.171	0.186	0.323	0.539	0.465	0.499
(f)	✗	✓	✓	✓	0.387	0.271	0.206	0.165	0.180	0.320	0.526	0.408	0.459
PriorRG	✓	✓	✓	✓	**0.412**	**0.290**	**0.220**	**0.175**	**0.189**	0.324	**0.541**	**0.485**	**0.511**

Table 2: Ablation study on MIMIC-CXR dataset. "CC" and "PI" denote clinical context and prior image, respectively.

Overview



Spatiotemporal Fusion Network

$$\boldsymbol{V}_{ca}^{st} = \mathrm{LN}(\boldsymbol{V}^{cur} + \mathrm{CA}(\mathrm{LN}(\boldsymbol{V}^{cur}), \mathrm{LN}(\boldsymbol{V}^{pri}))), \quad (1)$$

$$\boldsymbol{V}^{st} = \mathrm{LN}(\boldsymbol{V}_{ca}^{st} + \mathrm{FFN}(\boldsymbol{V}_{ca}^{st})), \quad (2)$$

Coarse-to-Fine Decoding

$$\bar{\boldsymbol{T}}^{c} = \mathrm{Perceiver}(\boldsymbol{E}^{lat}, \boldsymbol{T}^{c}), \quad (3)$$

$$\bar{\boldsymbol{V}}^{st} = \mathrm{Perceiver}(\bar{\boldsymbol{T}}^{c}, \boldsymbol{V}^{st}), \quad (4)$$

$$\bar{\boldsymbol{V}}^{hier} = \mathrm{Perceiver}(\bar{\boldsymbol{V}}^{st}, \boldsymbol{V}^{hier}). \quad (8)$$

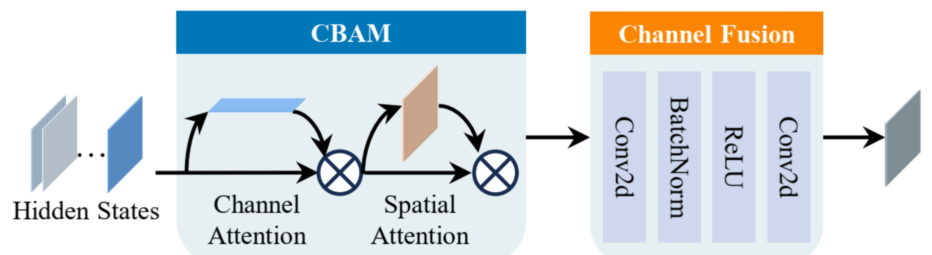
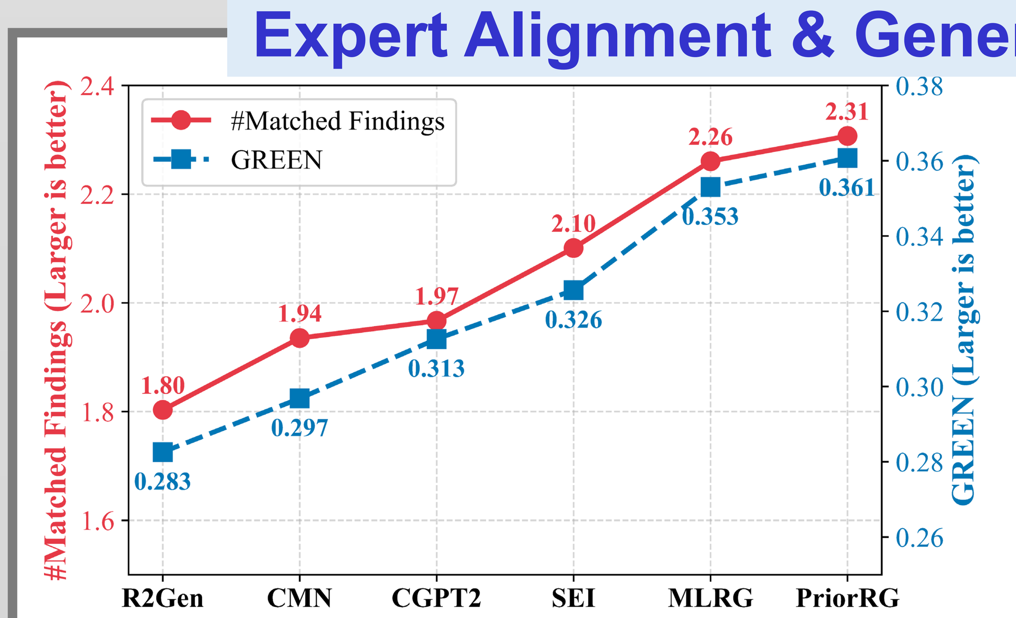


Figure 3: Overview of attention-enhanced layer fusion network (ALF), which leverages CBAM (Woo et al. 2018) to extract hierarchical visual features.

Expert Alignment & Generalization Capability



Method	Venu	BLEU-4↑	METEOR↑
Zero-shot Report Generation			
R2GenGPT	Meta-Radio'23	0.112	0.177
Med-LLM	MM'24	0.125	0.185
PriorRG (Ours)	-	**0.178**	**0.211**
Supervised Report Generation			
METransformer	CVPR'23	0.172	0.192
MAN	AAAI'24	0.170	0.213
DCG	MM'24	0.186	0.211
Med-LLM	MM'24	0.168	0.209
R2-LLM	AAAI'24	0.184	0.208
PriorRG (Ours)	-	**0.196**	**0.216**

Table 5: Comparison of report generation results on the IU X-ray dataset (Demner-Fushman et al. 2016) under zero-shot and supervised settings. **Best** results are in **Bold**.

Effect of Prior-Knowledge Ratio and Progressive Fusion

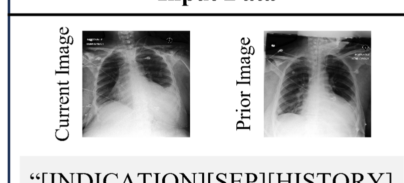
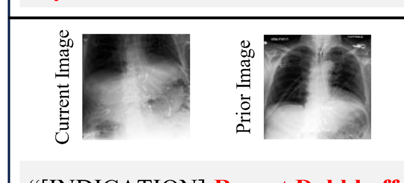
Prior Type	Ratio	B-2↑	F1↑
Prior Image (PI)	0%	0.287	0.513
	10%	0.287	0.513
	30%	0.288	0.513
	50%	**0.289**	**0.515**
	100%	**0.289**	**0.515**
Clinical Context (CC)	0%	0.139	0.492
	10%	0.157	0.495
	30%	0.187	0.498
	50%	0.220	0.501
	100%	**0.294**	**0.507**

Table 9: Report generation performance on the MIMIC-CXR dataset under varying proportions of Prior Image (PI) and Clinical Context (CC).

Model	NLG Metrics ↑ B-2	B-4	MTR	CE Metrics ↑ P	R	F1
LastOnly	0.278	0.168	0.185	**0.554**	0.466	0.506
Fine2coarse	0.284	0.172	0.187	0.549	0.482	**0.514**
PriorRG	**0.290**	**0.175**	**0.189**	0.541	**0.485**	0.511

Table 3: Comparison of different progressive fusion strategies for report generation on the MIMIC-CXR dataset.

Qualitative Analysis

Input Data	Radiologist-generated Report	AI-generated Report by SEI	AI-generated Report by PriorRG (Ours)
 Current Image, Prior Image "[INDICATION][SEP][HISTORY] ___-year-old woman with **respiratory failure**. Evaluate fluid status."	Comparison is made to the prior study from ___. The feeding tube, left IJ catheter and endotracheal tube are unchanged in position. There is persistent cardiomegaly. There is unchanged left retrocardiac opacity. There are no signs for overt pulmonary edema. <u>There is a small right-sided pleural effusion as well</u>. Overall, these findings are stable.	in comparison with the earlier study of this date, there has been placement of an endotracheal tube with its tip approximately 5 cm above the carina. otherwise, little change.	Comparison is made to prior study from ___. The **endotracheal tube, feeding tube, and left IJ central line are unchanged in position**. There is **unchanged cardiomegaly**. There is **a persistent left retrocardiac opacity** and left-sided pleural effusion which is stable. There are no signs for overt pulmonary edema. No pneumothoraces are identified.
 Current Image, Prior Image "[INDICATION] **Recent Dobbhoff tube placement**.[SEP][HISTORY]"	As compared to the previous radiograph, the Dobbhoff tube shows now normal course. The tip projects over the middle parts of the stomach. No complications, notably no pneumothorax. Otherwise, the image is unchanged.	as compared to the previous radiograph, the dobbhoff tube has been repositioned. the course of the tube is unremarkable, the tip is not included on the image. otherwise, there is no relevant change.	As compared to the previous radiograph, <u>the patient has received a</u> Dobbhoff catheter. The course of the catheter is unremarkable, the tip of the catheter projects over the middle parts of the stomach. There is no evidence of complications, notably no pneumothorax. Otherwise, the radiograph is unchanged.